
Cryptographic Authority Governance for Autonomous AI Agents

RFC-ATF-1: Agent Trust Fabric Protocol

Harold Alberto Nunes Rodelo
OMNIX QUANTUM LTD · Dubai, United Arab Emirates
harold@omnixquantum.com

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ML-DSA-65 (Dilithium-3, NIST FIPS 204)

Abstract

Autonomous AI agents increasingly make consequential decisions in enterprise systems — executing trades, authorizing medical procedures, routing logistics, and managing critical infrastructure. Yet the governance of *who authorized these agents, under what authority, and whether that authority was valid at the moment of execution* remains largely unaddressed by existing frameworks.

This paper presents the OMNIX Agent Trust Fabric (ATF) — a formally specified, post-quantum cryptographic protocol that answers four questions for every AI agent execution:

- Who authorized this agent?** — Via ML-DSA-65 signed Delegation Receipt (DR)
- What authority did it hold?** — Via Monotonic Authority Reduction invariant (MAR), model-checked in TLA+
- Was the authority valid at execution?** — Via Temporal Admissibility Record (TAR), captured before the pipeline runs

4. **Is the full chain independently verifiable?** — Via offline CLI verifier, no platform access required (ATF-INV-006)

The result is a three-artifact audit chain — DR + TAR + GovernanceReceipt — for every governance decision. This chain is PQC-signed, independently verifiable, and designed to satisfy regulatory accountability requirements under the EU AI Act, DORA, MiCA, and NIST AI RMF.

1. Problem Statement

1.1 The AI Agent Authorization Gap

Modern AI governance frameworks address *what decisions are made* but not *who authorized the agent that made them*. When an enterprise AI agent executes a high-stakes action, the audit record typically captures the decision, the signals that influenced it, and a hash of the receipt. What is missing:

- Proof that the agent was authorized to act — not just authenticated, but explicitly authorized with bounded scope
- Proof that the authority was valid at the nanosecond of execution — valid then, not valid in principle
- A verifiable chain from the AI agent back to a human principal — traceable without trusting the platform

1.2 The Accountability Requirement

Regulation is closing this gap:

Regulation	Requirement	ATF Response
EU AI Act Art. 13	Transparency — who authorized this system	ATF DR chain to Tier-1 human
EU AI Act Art. 14	Human oversight — human principal identifiable	ATF chain_root_id → Tier-1
DORA Art. 9	ICT resilience — documented controls for automated systems	TARs proves control at execution
MiCA Rec. 65	Algorithmic trading — authorization audit trail	ATF triple chain for trading decisions
NIST AI RMF	Accountable AI — decisions traceable to entities	Cryptographic traceability
SOC 2 CC6	Logical access controls — non-repudiation	ATF PQC signatures

2. Protocol Architecture

2.1 Core Artifacts

Delegation Receipt (ATFDR-`{16HEX}`): A PQC-signed record granting a bounded subset of the principal's authority for a defined task scope. Signed with ML-DSA-65, content-hashed (SHA-256 over canonical JSON), independently verifiable offline.

Temporal Admissibility Record (ATFTAR-`{16HEX}`): Issued at the exact moment of admission to the governance pipeline — before any governance logic executes. Proves the DR was ACTIVE at execution_ns and binds to a specific GovernanceReceipt via execution_ref.

GovernanceReceipt: OMNIX's existing governance decision record (ADR-028), extended with `atf_context` embedding the DR, TAR, and trust summary. Enables answering all four questions from a single JSON document.

2.2 Authority Budget Arithmetic

The authority budget is a real number in `[0.0, 100.0]`. The Monotonic Authority Reduction (MAR) invariant requires:

```
For all DRs D in the trust lattice: D.authority_budget_granted ≤
D.authority_budget_delegator
```

This invariant is formally specified in TLA+ and verified by model checking (ATF-FV-1.0). Delegation cannot create authority — only distribute and bound it. Any attempt to issue a DR with `budget_granted > budget_delegator` raises a `MARViolationError` and is rejected before signing.

3. Formal Invariants

Invariant	ID	Statement
Monotonic Authority Reduction	ATF-INV-001	<code>DR.budget_granted ≤ DR.budget_delegator</code> for all DRs
Acyclicity	ATF-INV-002	Trust lattice is a DAG — no delegation cycle exists
Chain Root Consistency	ATF-INV-003	All DRs in a chain share the same <code>chain_root_id</code>
Content Hash Immutability	ATF-INV-004	DR fields are immutable post-issuance
Temporal Non-Future-Dating	ATF-INV-005	<code>TAR.execution_ns ≤ current time at verification</code>
Independent Verifiability	ATF-INV-006	Full chain verifiable offline, no platform access required

TLA+ Coverage: INV-001 (MARInvariant), INV-002 (AcyclicityInvariant), INV-003 (ChainRootConsistency), INV-004 (ImmutabilityProperty) are formally specified and model-checked in `ATF-TLA-SPEC.tla`.

4. Cryptographic Specification

Property	Value
Algorithm	ML-DSA-65 (Dilithium-3)
Standard	NIST FIPS 204 (August 2024)
Security level	NIST PQC Level 3
Public key size	1952 bytes
Signature size	3293 bytes
Hash function	SHA-256 (content hash construction)
Encoding	Base64 (signature), hex (content hash)

Note: The reference implementation uses the `pqc` Python library (Dilithium-3). This library implements the ML-DSA-65 algorithm as specified in NIST FIPS 204 but has not undergone FIPS 140-3 module validation. For regulated deployments requiring FIPS 140-3, substitute a validated

cryptographic module.

5. Temporal Admissibility Protocol (ADR-157)

The Temporal Admissibility Record (TAR) makes the authority check explicit and produces a signed, persistent record of admission before any governance logic runs:

Property	JWT exp check	ATF TAR
Existence check	Yes	Yes
Signed artifact	No	Yes (ML-DSA-65)
Timestamp granularity	Second	Nanosecond-resolution
Persistence	No	Yes (database row)
Binding to execution	No	Yes (execution_ref)
Independent verifiability	No	Yes (offline CLI)
Rejection record	No	Yes (REJECTED status + reason)

6. Threat Model Summary

The full threat model (OMNIX-TM-ATF-2026-001) analyzes nine threat classes. Summary of residual risks:

Threat	Severity	Primary Defense	Residual Risk
Receipt Replay (TM-001)	HIGH	TAR uniqueness + execution_ref binding	LOW
DR Forgery (TM-002)	CRITICAL	ML-DSA-65 PQC signature	VERY LOW
Privilege Amplification (TM-003)	CRITICAL	MAR invariant + TLA+ verification	LOW
Clock Drift (TM-004)	MEDIUM	execution_ns in signed content hash	MEDIUM*
Chain Poisoning (TM-005)	HIGH	chain_root_id + verify_chain()	LOW
Orphan Delegation (TM-006)	HIGH	Cascade revocation + TAR check	MEDIUM**
DAG Cycle Injection (TM-007)	HIGH	TLA+ AcyclicityInvariant	VERY LOW
Cross-TAR Forgery (TM-008)	HIGH	execution_ref binding in signed hash	VERY LOW
Budget Inflation DTR (TM-009)	HIGH	CDTP-INV-003 + PQC-signed DTR	LOW

** On-premise deployments. Mitigated with RFC 3161 TSA. ** In distributed deployments without shared revocation registry.*

7. Compliance Framework

ATF defines three compliance levels:

- Level 1 — Basic (Development and Proof-of-Concept):** DR content hash, MAR enforcement, chain traversal. PQC signature SHOULD. TAR OPTIONAL.
- Level 2 — Standard (Production, Enterprise):** All Level-1 requirements. PQC signature REQUIRED. TAR REQUIRED. ATF CCS ≥ 80 for high-assurance domains. Independent verifiability demonstrated.

Level 3 — Sovereign (Regulated, Critical Infrastructure): All Level-2 requirements. RFC 3161 TSA counter-signature on TARs. FIPS 140-3 validated module SHOULD. Formal verification documentation. Public CLI verifier for independent audit.

8. Independent Verifiability (ATF-INV-006)

ATF-INV-006 is the most operationally significant invariant for regulators: any party **MUST** be able to verify a delegation chain using only the receipts and the root public key. No access to the issuing platform, API, account, or internet connection is required.

Public CLI verifier usage:

```
python omnix_atf_verify.py --mode receipt receipt.json
```

A regulator conducting a post-incident review downloads the receipt artifacts from the institution's disclosure, runs the verifier, and receives a complete VERIFIED/NOT VERIFIED verdict with full chain analysis — without any interaction with OMNIX or the deploying institution's systems.

9. Conclusion

The OMNIX Agent Trust Fabric addresses a specific, well-defined gap in AI governance infrastructure: the absence of cryptographic proof of who authorized an AI agent, under what authority bound, when that authority was valid, and whether execution was admitted at that exact moment.

ATF provides this through a three-artifact chain (DR + TAR + GovernanceReceipt), formally specified invariants (TLA+ model-checked), post-quantum cryptography (ML-DSA-65, NIST FIPS 204), and independent offline verifiability (ATF-INV-006). The design is complementary to existing orchestration frameworks — it operates at the governance layer that precedes execution.

As autonomous AI systems take on greater responsibility in enterprise and regulated environments, this governance layer becomes a prerequisite for institutional trust. OMNIX ATF provides the formal specification, reference implementation, and public verification tooling to establish that trust on a cryptographic foundation.

References

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